PARISHRAM 2.0

CLASS 12TH 2026

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-4

-PARAM SIR



TODAY'S TOPIC -

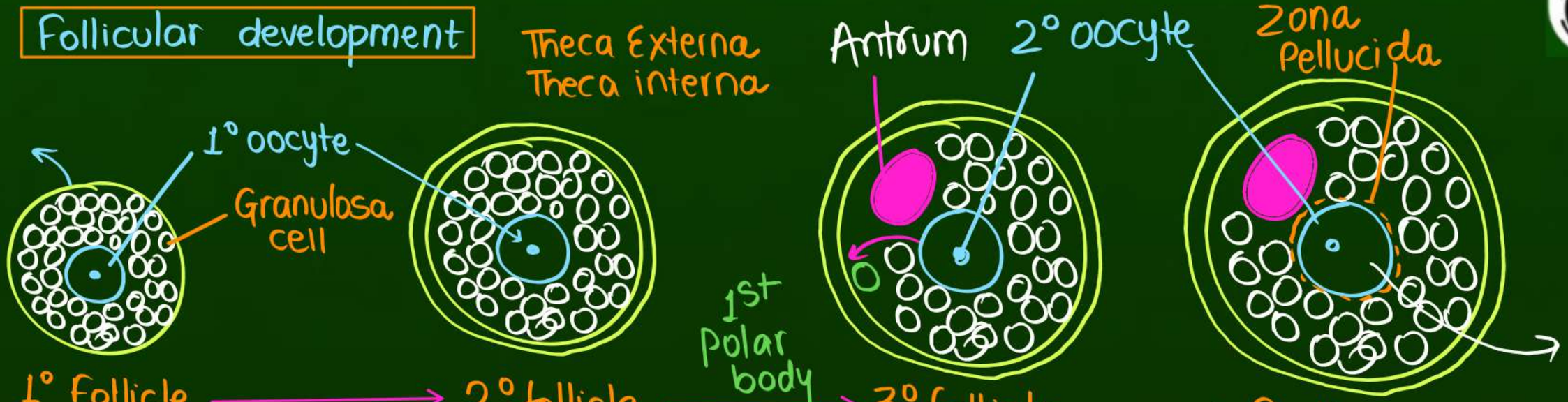
1. REVISION
2. OOGENESIS
3. TYPE OF FOLLICLES



OOGENESIS & TYPE OF FOLLICLES

- ✓ Formation of OVA | OVUM | EGG | ♀ Gamete
- ✓ Formation of mature ♀ gamete
- ✓ Process Starts during Embryonic life

Follicular development



1° Follicle → 2° follicle → 3° follicle → Graffin follicle

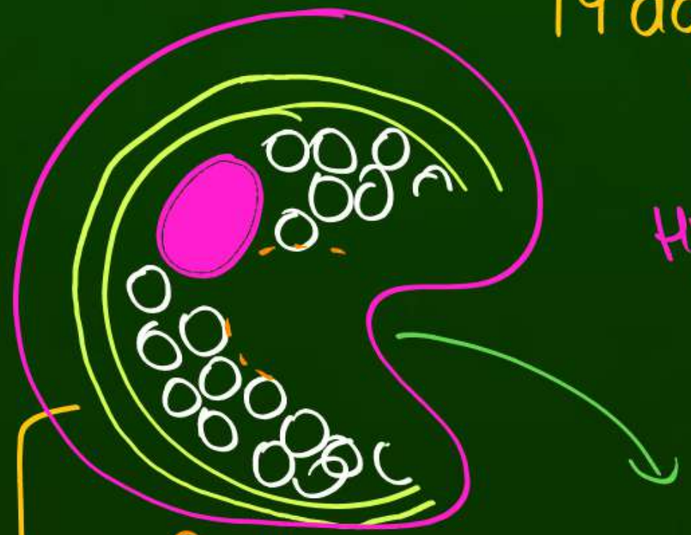
① 1° oocyte
② Granulosa Cell ↑
③ During Gestation Period

① 1° oocyte
② Granulosa Cell ↑↑ ★
③ Theca Externa + Interna

① 2° oocyte ★
② Granulosa Cell ↑↑↑
③ Theca Externa + Interna
④ Meiosis 1 Complete
⑤ Antrum Present ★

① 2° oocyte
② GC ↑↑↑↑
③ Theca E + I
④ Antrum ⊕
5 Zona Pellucida ★

14 day of Menstrual Cycle



Graffin follicle

Courpus luteum



Hyaluronic acid

cumulus

corona radiata

zona pellucida

2° oocyte

2° oocyte | ovum | Mature ♀ Gamete | Egg

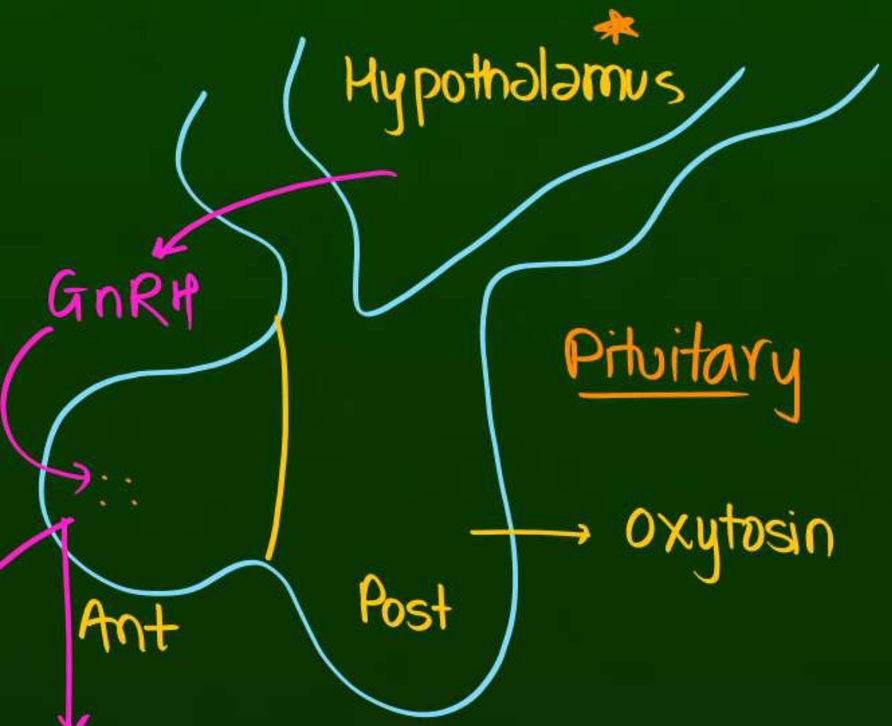
Corpus luteum

Some amount of Estrogen

PROGESTRON → Maintain Endometrium

Hormonal changes

[Gonadotropin Regulating Hormone]



① → ovulation
② → LH surge
③ → formation of curpus lutuem

→ Follicular development
→ Formation of Granulosa cell → Estrogen

EXTRACT

The process of formation of a mature female gamete is called **oogenesis** which is markedly different from spermatogenesis. Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (**oogonia**) are formed within each fetal ovary; no more oogonia are formed and added after birth. These cells start division and enter into prophase-I of the meiotic division and get temporarily arrested at that stage, called **primary oocytes**. Each primary oocyte then gets surrounded by a layer of granulosa cells and is called the **primary follicle** (Figure 2.7). A large number of these follicles degenerate during the phase from birth to puberty. Therefore, at puberty only 60,000-80,000 primary follicles are left in each ovary. The primary follicles get surrounded by more layers of granulosa cells and a new theca and are called **secondary follicles**.

The secondary follicle soon transforms into a tertiary follicle which is characterised by a fluid filled cavity called **antrum**. The theca layer is organised into an inner theca interna and an outer theca externa. It is important to draw your attention that it is at this stage that the primary oocyte within the tertiary follicle grows in size and completes its first meiotic division. It is an unequal division resulting in the formation of a large haploid **secondary oocyte** and a tiny first polar body (Figure 2.8b). The

NCERT

EXTRACT

secondary oocyte retains bulk of the nutrient rich cytoplasm of the primary oocyte. *Can you think of any advantage for this?* Does the first polar body born out of first meiotic division divide further or degenerate? At present we are not very certain about this. The tertiary follicle further changes into the mature follicle or **Graafian follicle** (Figure 2.7). The secondary oocyte forms a new membrane called **zona pellucida** surrounding it. The Graafian follicle now ruptures to release the secondary oocyte (ovum) from the ovary by the process called **ovulation**. *Can you identify major differences between spermatogenesis and oogenesis?* A di

NCERT-dig

Arrested Prophase I → Diplotene

FSH

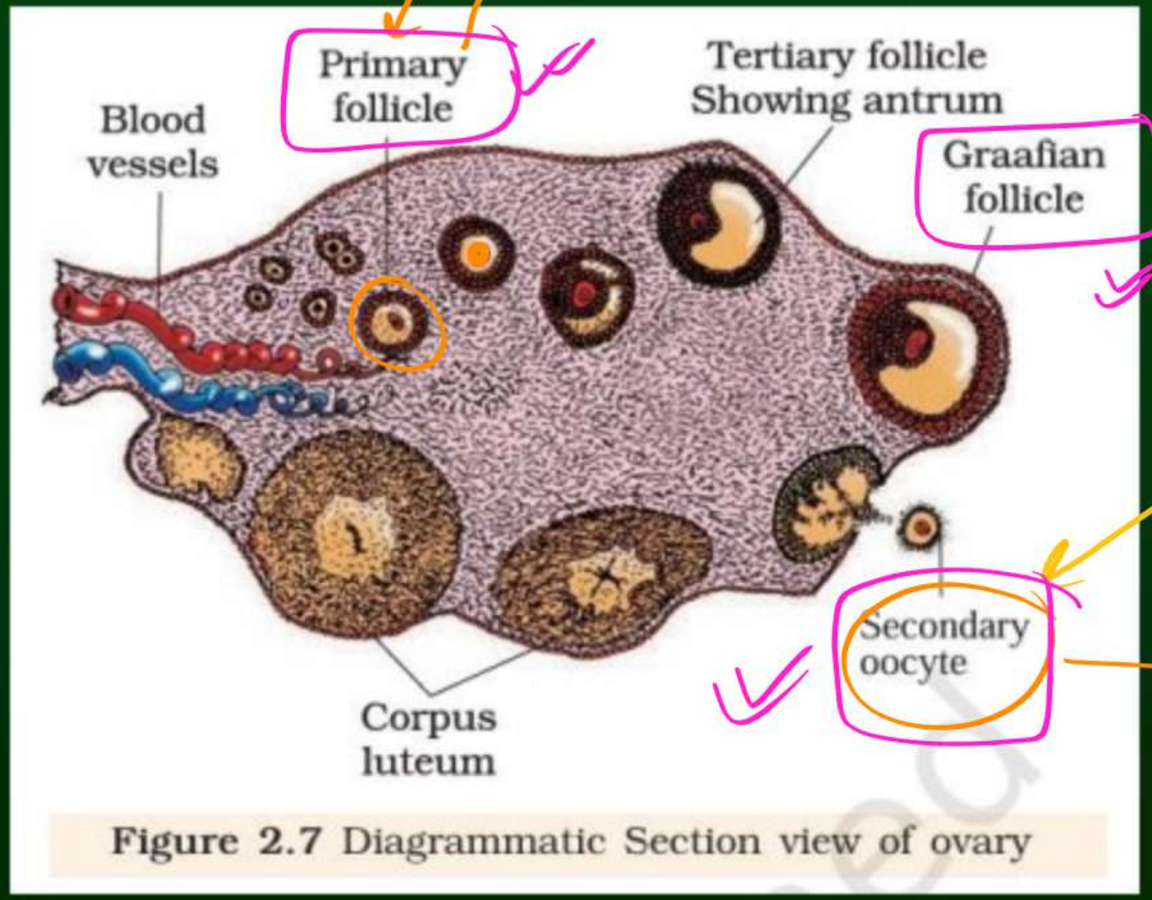


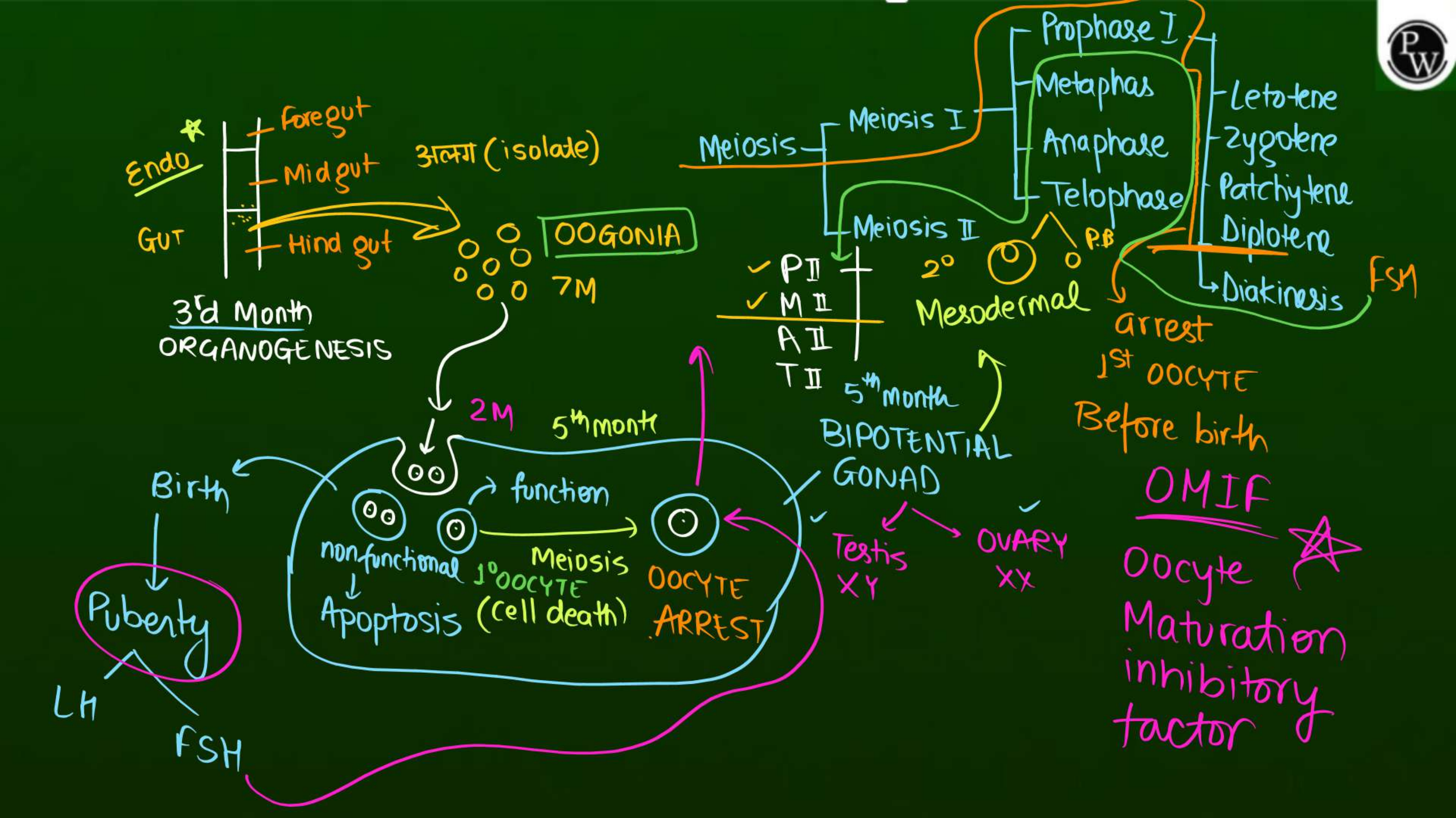
Figure 2.7 Diagrammatic Section view of ovary

Meiosis II

Metaphase II

Arrested

→ ovum / ♀ gamete


Endo
Foregut
Midgut
Hind gut
GUT
अलग (isolate)
OOGONIA
7M
3rd Month
ORGANOGENESIS
2M
5th month
Birth
Puberty
LH
FSH
function
non functional
Apoptosis
1° OOCYTE
Meiosis
(cell death)
OOCYTE
ARREST
Meiosis
Meiosis I
Prophase I
Metaphas
Anaphase
Telophase
Leto-tene
Zygotene
Patchytene
Diplotene
Diakinesis
FSH
Meiosis II
PII
MII
AII
TII
2°
P.B
Mesodermal
arrest
1st OOCYTE
Before birth
5th month
BIPOTENTIAL
GONAD
Testis
XY
OVARY
XX
OMIF
Oocyte
Maturation
inhibitory
factor

OOGONIA

↓ Mitosis

1° OOCYTE

Before birth

Before birth

[arrested PROPHASE I : Diplotene Stage]

Purberty

Meiosis I complete at 12th day of M.C

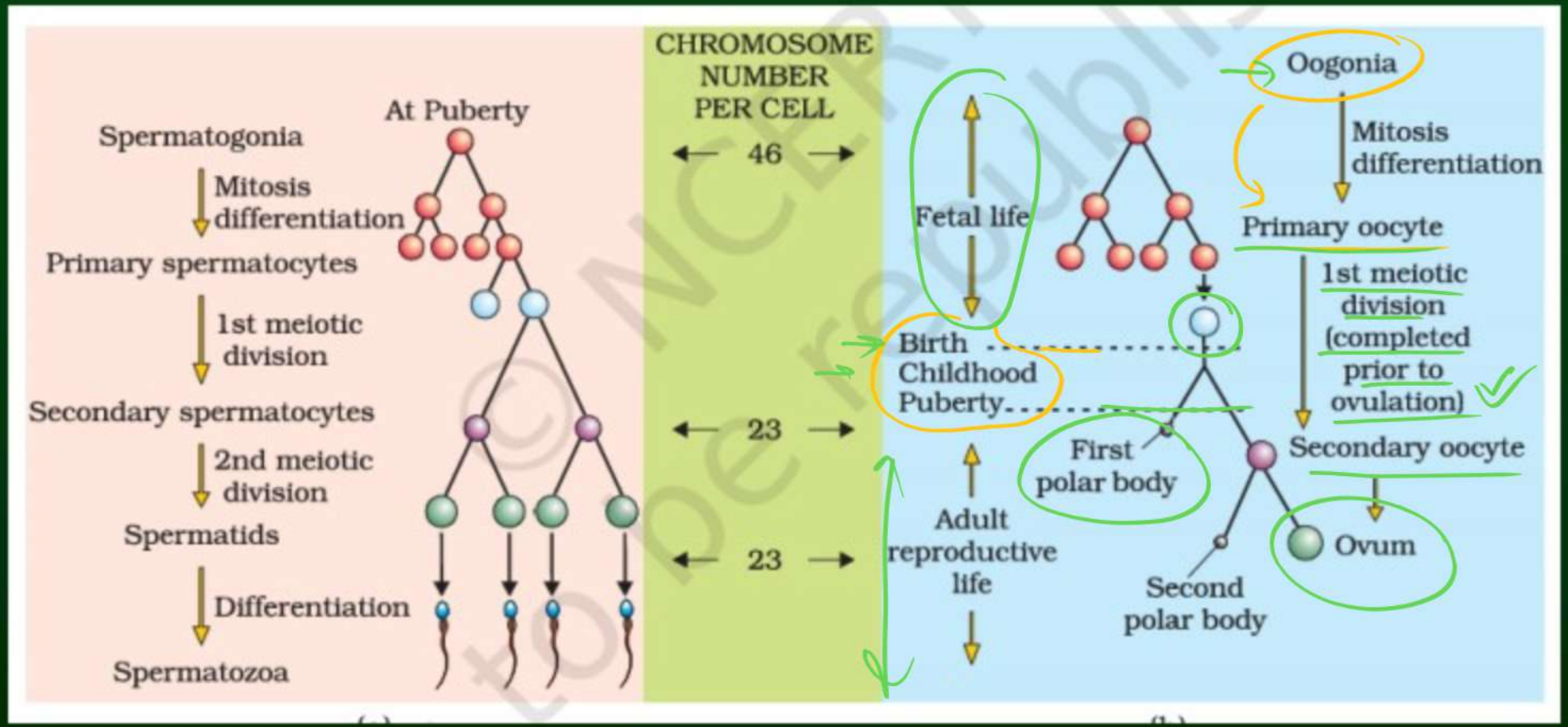
2° oocyte

Polar body

arrested : Metaphase II

→ Meiosis complete after fertilization

NCERT-dig


At Puberty
Spermatogonia
Mitosis differentiation
Primary spermatocytes
1st meiotic division
Secondary spermatocytes
2nd meiotic division
Spermatids
Differentiation
Spermatozoa
CHROMOSOME NUMBER PER CELL
46
23
23
Fetal life
Birth
Childhood
Puberty
Adult reproductive life
Oogonia
Mitosis differentiation
Primary oocyte
1st meiotic division (completed prior to ovulation)
Secondary oocyte
First polar body
Second polar body
Ovum

QUESTION

Which among the following has 23 chromosomes?

1. Spermatogonia and oogonia
2. Primary spermatocyte and primary oocyte
3. Primary spermatocyte and secondary oocyte
4. Secondary spermatocyte and secondary oocyte

